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328672 (28)

BE (6th Semester)
Examination, April - May, 2016

(New Scheme)

Operating System

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Time Allowed : 3 hours *Maximum Marks* : 80
Minimum Pass Marks : 28

Note : (*i*) Part (*a*) of each question is compulsory and solve any two parts from (*b*), (*c*) and (*d*).

(*ii*) The figures in the right-hand margin indicate marks.

Unit - I

1. (*a*) What is an operating system? 2

(*b*) Why is the operating system viewed as a resource allocator and control program? 7

(*c*) Explain real-time systems with examples of at least two applications. 7

(*d*) Explain the types of directory structure. 7

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Unit - II

2. (*a*) What is 'response time'? 2

(*b*) What is a process state? Mention the various states of a process. 7

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(*c*) Explain the various CPU scheduling algorithms. 7

(*d*) Consider the situation in which the disk read/write head is currently located at track 45 (of tracks 0-255) and moving in the positive direction. Assume that the following track requests have been made in this order : 40, 67, 11, 240, 87. What is the order in which optimized C-SCAN would service these requests and what is the total seek distance? 7

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Unit - III

3. (*a*) Name two differences between logical and physical address. 2

(*b*) Explain the difference between internal and external fragmentation. 7

(*c*) List the steps needed to perform page replacement. Explain the different page replacement policies. List out the main requirements, which should be satisfied by a page replacement policy. 7

(*d*) Explain memory management using segmentation with paging. 7

Unit - IV

4. (*a*) What is safe state and unsafe state in operating system? 2

(*b*) List and explain the three events concerning resource allocation. Define the following : 7

(*i*) Resource request and allocation graph (RRAG)

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(*ii*) Wait for graph (WFG)

(*c*) Define deadlock. Explain the necessary condition for deadlock to occur. 7

(*d*) What is critical section problem? Give two solutions for critical section problem. 7

Unit - V

5. (*a*) What is block device? 2

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(*b*) Explain the goals of I/O software. 7

(*c*) Write short notes on : 7

(*i*) Interrupt Handlers

(*ii*) Device Drivers

(*d*) Explain direct memory access DMA. 7

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